SKTC 1623

Thermodynamics

Chapter 10: Vapor and Combined Power Cycles

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10.1 The Carnot Vapor Cycle

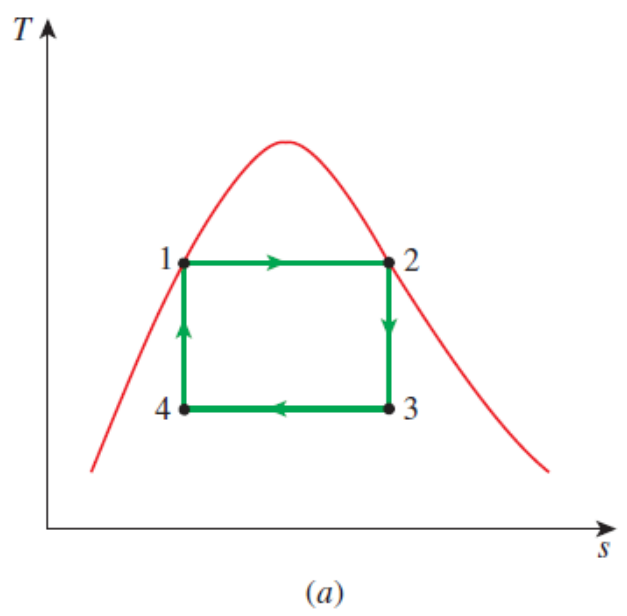


FIGURE 10–1
T-s diagram of two Carnot vapor cycles.

Process	Description
1-2	isothermal heat addition in a boiler
2-3	isentropic expansion in a turbine
3-4	isothermal heat rejection in a condenser
4-1	isentropic compression in a compressor

- The Carnot cycle is the most efficient cycle operating between two specified temperature limits but it is not a suitable model for vapor power cycles, since:
  - **Process 1-2** Limiting the heat transfer processes to two-phase systems severely limits the maximum temperature that can be used in the cycle (374°C for water)
  - **Process 2-3** The turbine cannot handle steam with a high moisture content because of the impingement of liquid droplets on the turbine blades causing erosion and wear.
    - Usually only with quality no less than 90%
  - **Process 4-1** It is not practical to design a compressor that handles two phases.

10.1 The Carnot Vapor Cycle (2)

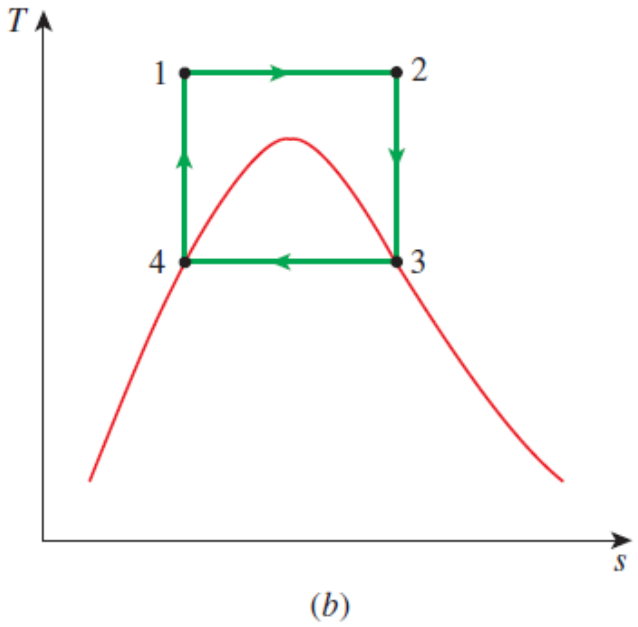


FIGURE 10–1
T-s diagram of two Carnot vapor cycles.

- The cycle in (b) could eliminate some of the previous problem but it is not suitable since it requires isentropic compression to extremely high pressures and isothermal heat transfer at variable pressures.
- Thus it can be concluded that the Carnot cycle can be approximated in actual devices and it is not a realistic model for vapor power cycles
- To resolve the difficulties associated with the Carnot cycle, the **Rankine cycle** was devised.

10.2 Rankine Cycle: The Ideal Cycle For Vapor Power Cycles

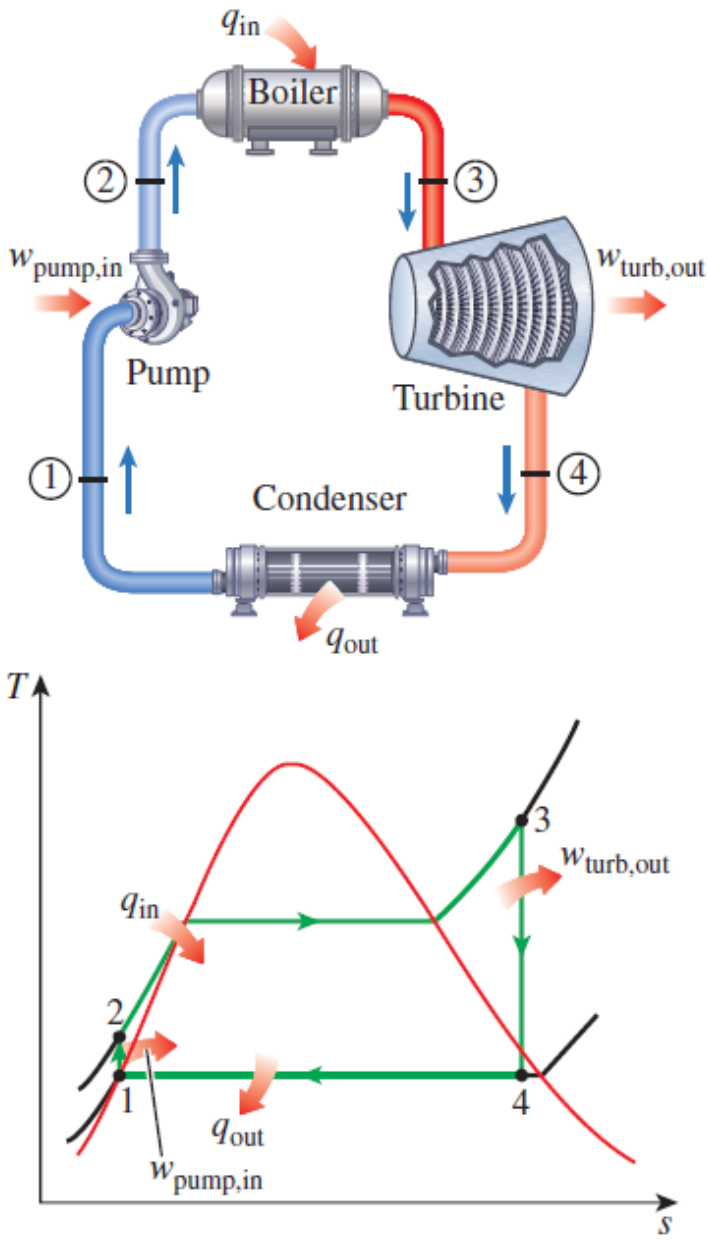


FIGURE 10–2
The simple ideal Rankine cycle.

- Many of the impracticalities associated with the Carnot cycle can be eliminated by superheating the steam in the boiler and condensing it completely in the condenser.
- The cycle that results is the Rankine cycle, which is the ideal cycle for vapor power plants.
- The ideal Rankine cycle does not involve any internal irreversibilities.

Process	Description
1-2	Isentropic Compression in a pump
2-3	Constant pressure Heat Addition in a boiler
3-4	Isentropic Expansion in a turbine
4-1	Constant pressure Heat Rejection in a condenser

10.2 Rankine Cycle: The Ideal Cycle For Vapor Power Cycles (2)

Energy Analysis of the Ideal Rankine Cycle

- All four components associated with the Rankine cycle (the pump, boiler, turbine, and condenser) are steady flow devices. Steady flow energy equation per unit mass is

$$(q_{in} - q_{out}) + (w_{in} - w_{out}) = h_e - h_i \quad \text{(kJ/kg)} \qquad (10.1)$$

- The boiler and condenser do not involve any work, and the pump and the turbine are assumed to be isentropic.
- The conservation of energy relation for each device can be expressed as follows:

$$\textit{Pump } (q = 0): \qquad w_{pump,in} = h_2 - h_1 \qquad (10.2)$$

- Recall the property relation: $dh = Tds + vdP$
- Since the ideal pumping process 1-2 is isentropic, $ds = 0$.

$$dh = vdP$$

$$\Delta h = h_2 - h_1 = \int_1^2 vdP \cong v_1 (P_2 - P_1)$$

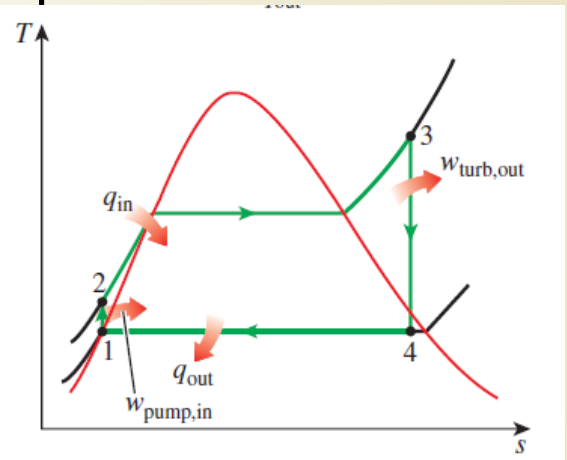


FIGURE 10–2
The simple ideal Rankine cycle.

10.2 Rankine Cycle: The Ideal Cycle For Vapor Power Cycles (3)

So $w_{pump,in} = h_2 - h_1$

or, $w_{pump,in} = v\,(P_2 - P_1)$ (10.3)

Where $h_1 = h_{f\,@\,P1}$ and $v \approx v_1 = v_{f\,@\,P1}$ (10.4)

Boiler (w = 0): $q_{in} = h_3 - h_2$ (10.5)

Turbine (q = 0) $w_{turb,out} = h_3 - h_4$ (10.6)

Condenser (w = 0): $q_{out} = h_4 - h_1$ (10.7)

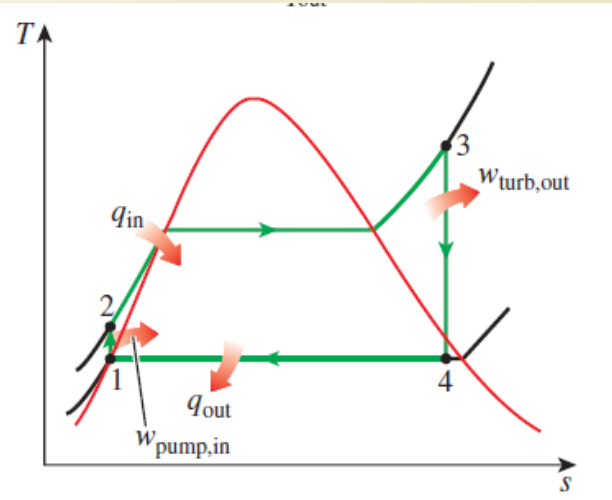


FIGURE 10–2
The simple ideal Rankine cycle.

- The *thermal efficiency* of the Rankine cycle is determined from

$$\eta_{th} = \frac{w_{net}}{q_{in}} = 1 - \frac{q_{out}}{q_{in}} \qquad (10.8)$$

where $w_{net} = q_{in} - q_{out} = w_{turb,out} - w_{pump,in}$

Example 1

EXAMPLE 10–1 The Simple Ideal Rankine Cycle

Consider a steam power plant operating on the simple ideal Rankine cycle. Steam enters the turbine at 3 MPa and 350°C and is condensed in the condenser at a pressure of 75 kPa. Determine the thermal efficiency of this cycle.

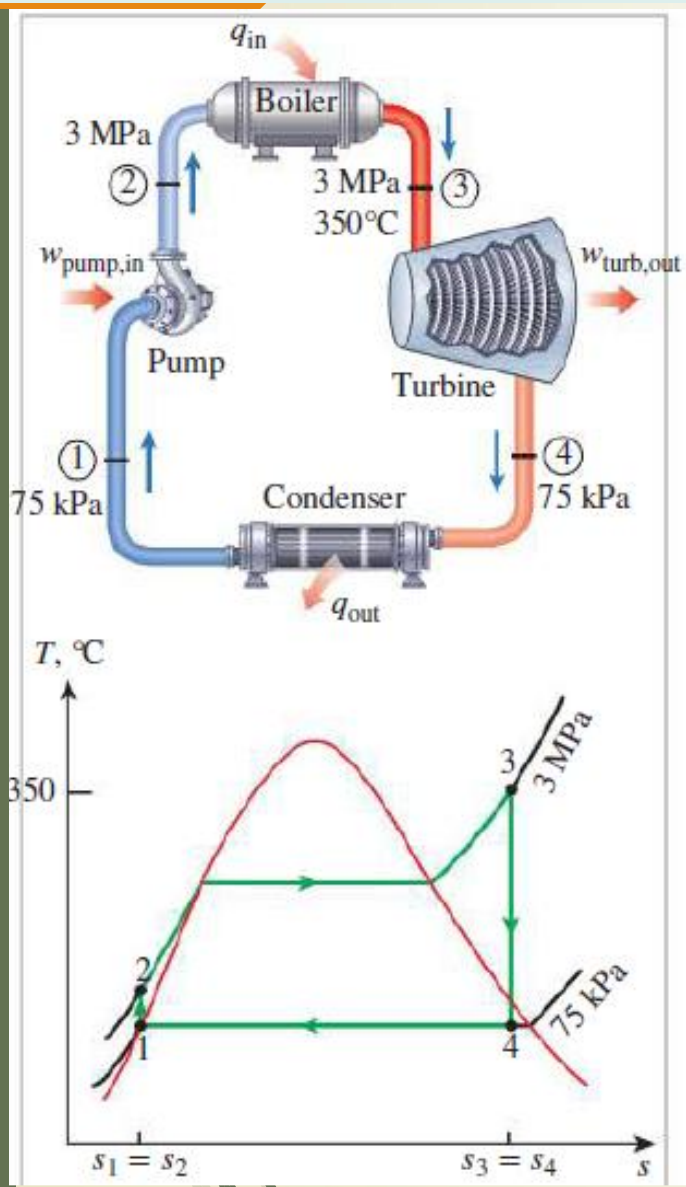


FIGURE 10–3
Schematic and *T-s* diagram for Example 10–1.

SOLUTION A steam power plant operating on the simple ideal Rankine cycle is considered. The thermal efficiency of the cycle is to be determined.

Assumptions **1** Steady operating conditions exist. **2** Kinetic and potential energy changes are negligible.

Analysis The schematic of the power plant and the *T-s* diagram of the cycle are shown in Fig. 10–3. We note that the power plant operates on the ideal Rankine cycle. Therefore, the pump and the turbine are isentropic, there are no pressure drops in the boiler and condenser, and steam leaves the condenser and enters the pump as saturated liquid at the condenser pressure.

First we determine the enthalpies at various points in the cycle, using data from steam tables (Tables A–4, A–5, and A–6):

State 1:

$$\left.\begin{array}{l} P_1 = 75 \text{ kPa} \\ \text{Sat. liquid} \end{array}\right\} \begin{array}{l} h_1 = h_{f\,@\,75\text{ kPa}} = 384.44 \text{ kJ/kg} \\ v_1 = v_{f\,@\,75\text{ kPa}} = 0.001037 \text{ m}^3\text{/kg} \end{array}$$

State 2: $P_2 = 3$ MPa

$$s_2 = s_1$$

$$w_{\text{pump,in}} = v_1(P_2 - P_1) = (0.001037 \text{ m}^3\text{/kg})[(3000 - 75) \text{ kPa}]\left(\frac{1 \text{ kJ}}{1 \text{ kPa}\cdot\text{m}^3}\right)$$

$$= 3.03 \text{ kJ/kg}$$

$$h_2 = h_1 + w_{\text{pump,in}} = (384.44 + 3.03) \text{ kJ/kg} = 387.47 \text{ kJ/kg}$$

State 3:

$$\left.\begin{array}{l} P_3 = 3 \text{ MPa} \\ T_3 = 350°\text{C} \end{array}\right\} \begin{array}{l} h_3 = 3116.1 \text{ kJ/kg} \\ s_3 = 6.7450 \text{ kJ/kg}\cdot\text{K} \end{array}$$

State 4: $P_4 = 75$ kPa (sat. mixture)

$$s_4 = s_3$$

$$x_4 = \frac{s_4 - s_f}{s_{fg}} = \frac{6.7450 - 1.2132}{6.2426} = 0.8861$$

$$h_4 = h_f + x_4 h_{fg} = 384.44 + 0.8861(2278.0) = 2403.0 \text{ kJ/kg}$$

Thus,

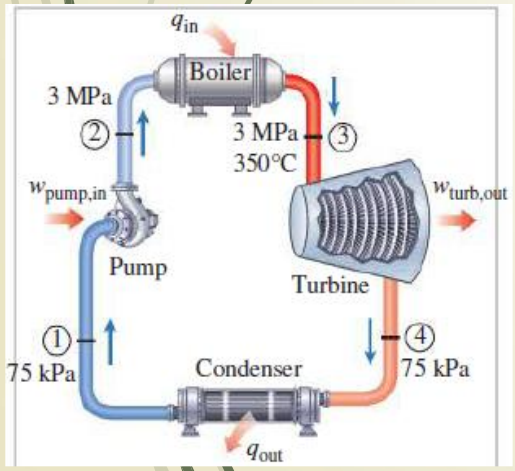
$$q_{in} = h_3 - h_2 = (3116.1 - 387.47)\ \text{kJ/kg} = 2728.6\ \text{kJ/kg}$$
$$q_{out} = h_4 - h_1 = (2403.0 - 384.44)\ \text{kJ/kg} = 2018.6\ \text{kJ/kg}$$

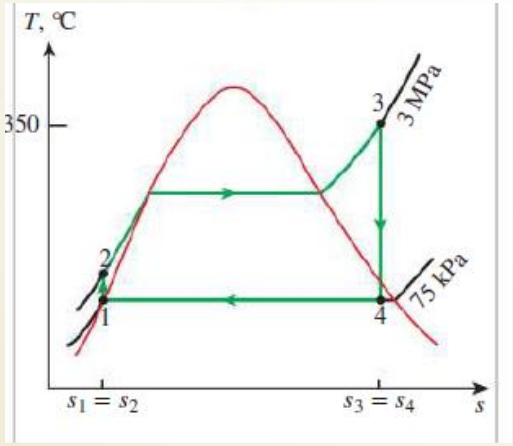
and

$$\eta_{th} = 1 - \frac{q_{out}}{q_{in}} = 1 - \frac{2018.6\ \text{kJ/kg}}{2728.6\ \text{kJ/kg}} = 0.260 \text{ or } \mathbf{26.0\%}$$

The thermal efficiency could also be determined from

$$w_{turb,out} = h_3 - h_4 = (3116.1 - 2403.0)\ \text{kJ/kg} = 713.1\ \text{kJ/kg}$$
$$w_{net} = w_{turb,out} - w_{pump,in} = (713.1 - 3.03)\ \text{kJ/kg} = 710.1\ \text{kJ/kg}$$





or

$$w_{net} = q_{in} - q_{out} = (2728.6 - 2018.6)\ \text{kJ/kg} = 710.0\ \text{kJ/kg}$$

and

$$\eta_{th} = \frac{w_{net}}{q_{in}} = \frac{710.0\ \text{kJ/kg}}{2728.6\ \text{kJ/kg}} = 0.260 \text{ or } \mathbf{26.0\%}$$

That is, this power plant converts 26 percent of the heat it receives in the boiler to net work. An actual power plant operating between the same temperature and pressure limits will have a lower efficiency because of the irreversibilities such as friction.

Discussion Notice that the back work ratio ($r_{bw} = w_{in}/w_{out}$) of this power plant is 0.004, and thus only 0.4 percent of the turbine work output is required to operate the pump. Having such low back work ratios is characteristic of vapor power cycles. This is in contrast to the gas power cycles, which typically involve very high back work ratios (about 40 to 80 percent).

It is also interesting to note the thermal efficiency of a Carnot cycle operating between the same temperature limits

$$\eta_{th,Carnot} = 1 - \frac{T_{min}}{T_{max}} = 1 - \frac{(91.76 + 273)\ \text{K}}{(350 + 273)\ \text{K}} = 0.415$$

Here T_{min} is taken as the saturation temperature of water at 75 kPa. The difference between the two efficiencies is due to the large external irreversibility in the Rankine cycle caused by the large temperature difference between steam and the heat source.

10.3 Deviation of Actual Vapor Power Cycles from Idealized Ones

- The actual vapor power cycle differs from the ideal Rankine cycle as a result of irreversibilities in various components.
- Fluid friction and heat loss to the surroundings are the two common sources of irreversibilities.

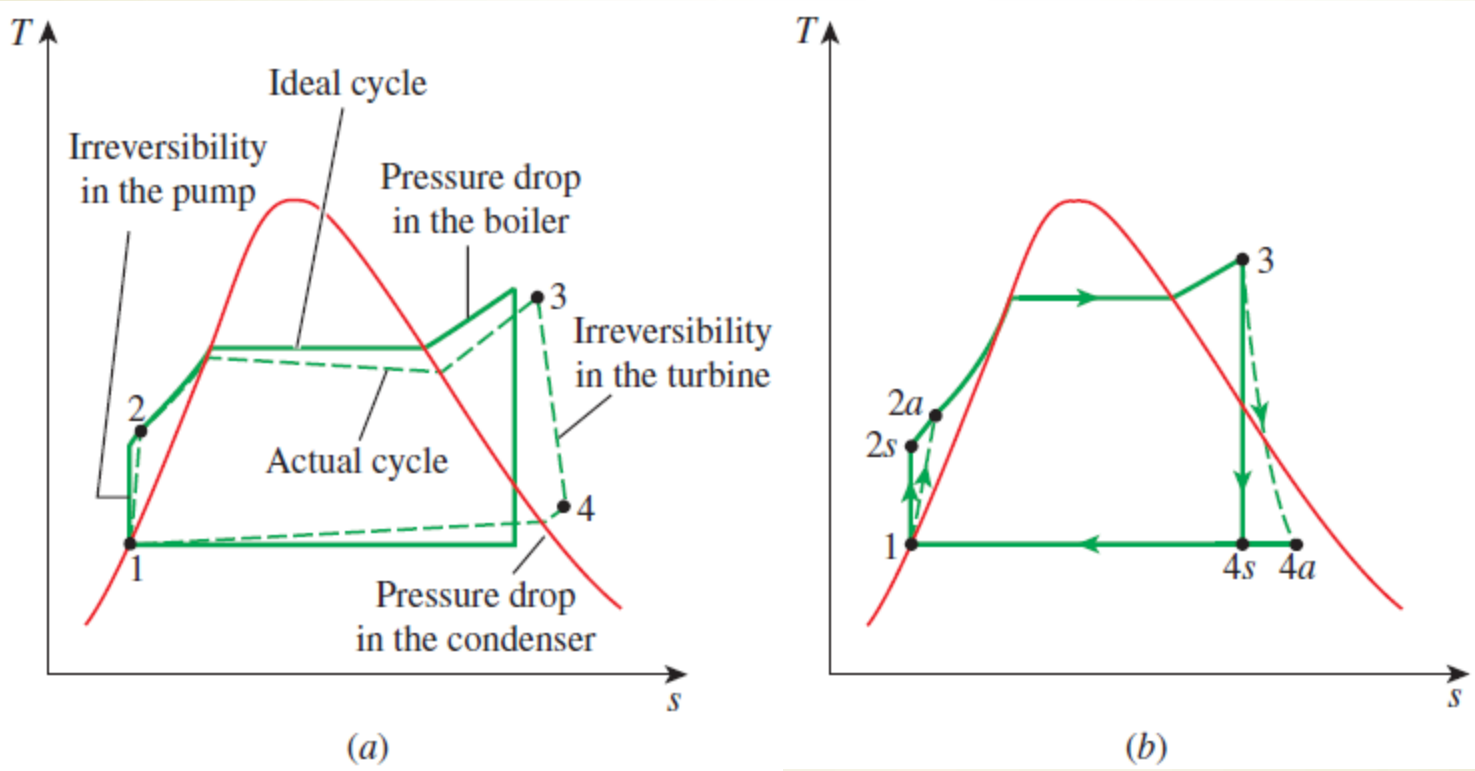


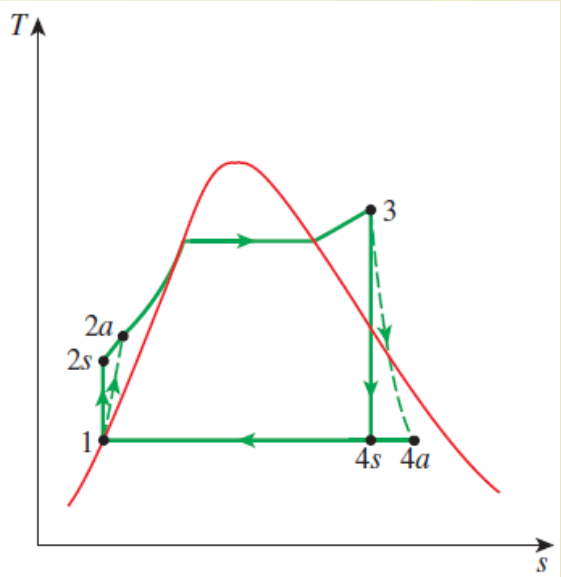
FIGURE 10–4
(*a*) Deviation of actual vapor power cycle from the ideal Rankine cycle. (*b*) The effect of pump and turbine irreversibilities on the ideal Rankine cycle.

10.3 Deviation of Actual Vapor Power Cycles from Idealized Ones (2)

- Fluid friction causes pressure drops in the boiler, the condenser, and the piping between various components.
- The major source of irreversibility is the heat loss from the steam to the surroundings as the steam flows through various components.
- To maintain the same level of net work output, more heat to be transferred to the steam in the boiler to compensate for these undesired heat losses.
- A pump requires a greater work input, and a turbine produces a smaller work output as a result of irreversibilities.

$$\eta_P = \frac{w_s}{w_a} = \frac{h_{2s} - h_1}{h_{2a} - h_1}; \quad and \quad \eta_T = \frac{w_a}{w_s} = \frac{h_3 - h_{4a}}{h_3 - h_{4s}}$$

Where states 2*a* and 4*a* are the actual and 2*s* and 4*s* are the isentropic exit states of the pump and turbine respectively.



(b)

Example 2

EXAMPLE 10–2 An Actual Steam Power Cycle

A steam power plant operates on the cycle shown in Fig. 10–5. If the isentropic efficiency of the turbine is 87 percent and the isentropic efficiency of the pump is 85 percent, determine (*a*) the thermal efficiency of the cycle and (*b*) the net power output of the plant for a mass flow rate of 15 kg/s.

SOLUTION A steam power cycle with specified turbine and pump efficiencies is considered. The thermal efficiency and the net power output are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Kinetic and potential energy changes are negligible.

Analysis The schematic of the power plant and the *T-s* diagram of the cycle are shown in Fig. 10–5. The temperatures and pressures of steam at various points are also indicated on the figure. We note that the power plant involves steady-flow components and operates on the Rankine cycle, but the imperfections at various components are accounted for.

(*a*) The thermal efficiency of a cycle is the ratio of the net work output to the heat input, and it is determined as follows:

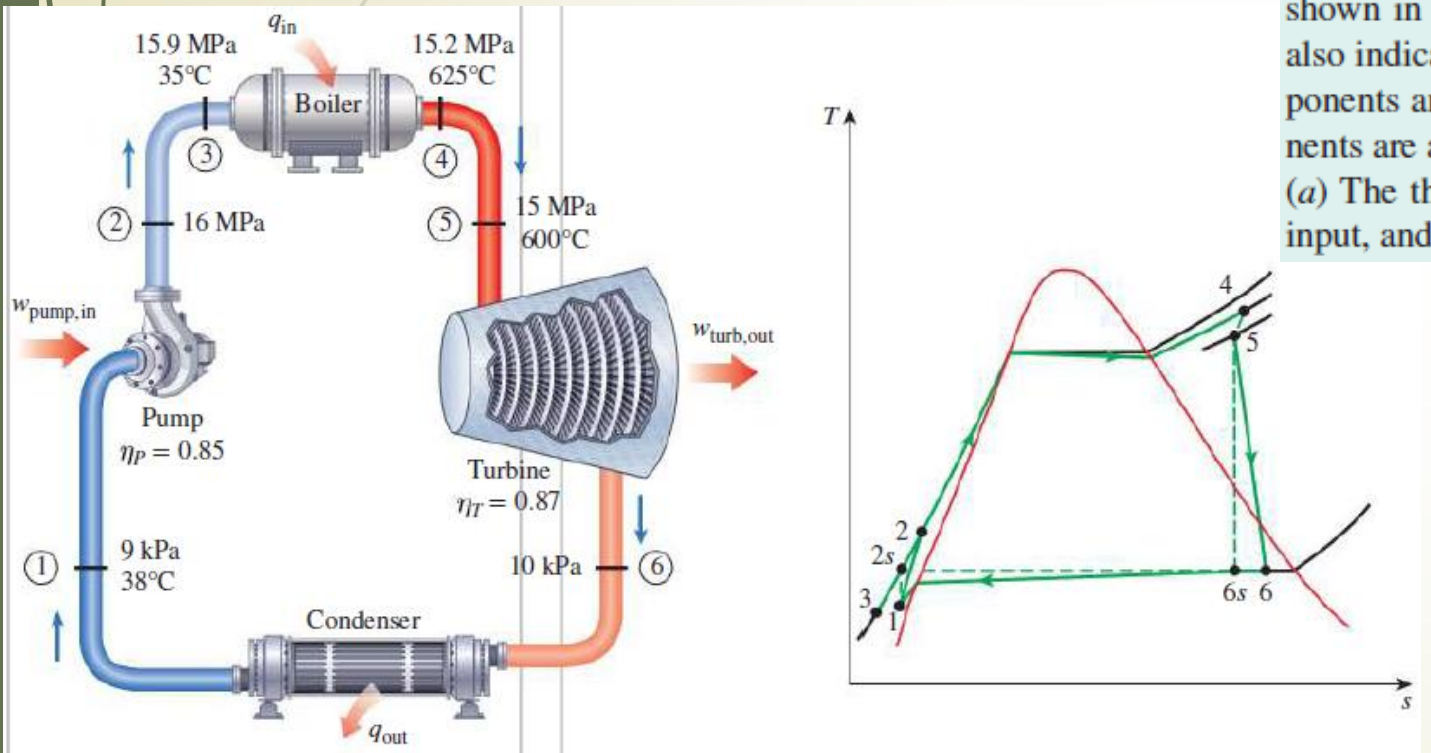


FIGURE 10–5
Schematic and *T-s* diagram for Example 10–2.

Pump work input:

$$w_{\text{pump,in}} = \frac{w_{s,\text{pump,in}}}{\eta_P} = \frac{v_1(P_2 - P_1)}{\eta_P}$$

$$= \frac{(0.001009\ \text{m}^3/\text{kg})[(16{,}000 - 9)\ \text{kPa}]}{0.85}\left(\frac{1\ \text{kJ}}{1\ \text{kPa}\cdot\text{m}^3}\right)$$

$$= 19.0\ \text{kJ/kg}$$

Turbine work output:

$$w_{\text{turb,out}} = \eta_T w_{s,\text{turb,out}}$$

$$= \eta_T(h_5 - h_{6s}) = 0.87(3583.1 - 2115.3)\ \text{kJ/kg}$$

$$= 1277.0\ \text{kJ/kg}$$

Boiler heat input: $q_{\text{in}} = h_4 - h_3 = (3647.6 - 160.1)\ \text{kJ/kg} = 3487.5\ \text{kJ/kg}$

Thus,

$$w_{\text{net}} = w_{\text{turb,out}} - w_{\text{pump,in}} = (1277.0 - 19.0)\ \text{kJ/kg} = 1258.0\ \text{kJ/kg}$$

$$\eta_{\text{th}} = \frac{w_{\text{net}}}{q_{\text{in}}} = \frac{1258.0\ \text{kJ/kg}}{3487.5\ \text{kJ/kg}} = 0.361 \text{ or } \mathbf{36.1\%}$$

(*b*) The power produced by this power plant is

$$\dot{W}_{\text{net}} = \dot{m}\, w_{\text{net}} = (15\ \text{kg/s})(1258.0\ \text{kJ/kg}) = \mathbf{18.9\ MW}$$

Discussion Without the irreversibilities, the thermal efficiency of this cycle would be 43.0 percent (see Example 10–3*c*).

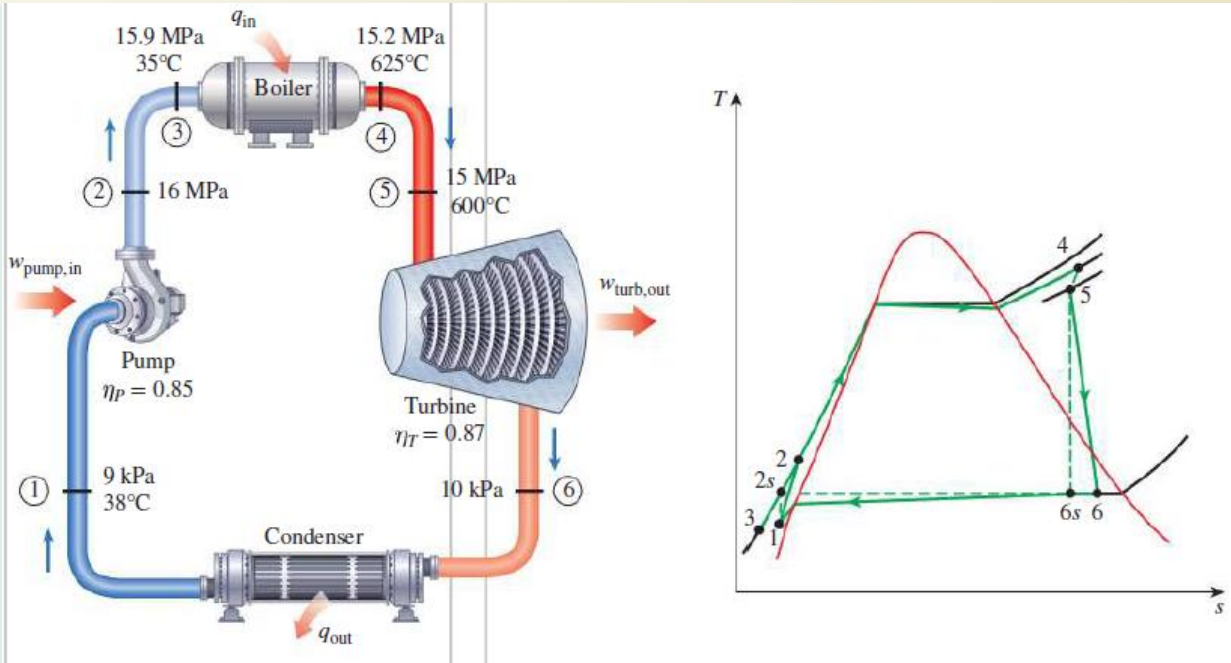


FIGURE 10–5
Schematic and *T-s* diagram for Example 10–2.

10.4 Ways to improve the simple Rankine cycle efficiency

- The basic idea behind all the modifications to increase the thermal efficiency of a power cycle is the same: Increase the average temperature at which heat is transferred to the working fluid in the boiler, or decrease the average temperature at which heat is rejected from the working fluid in the condenser.

1. **Lowering the condenser pressure (*Lowers* $T_{low,avg}$)**
   - To take advantage of the increased efficiencies at low pressures, the condensers of steam power plants usually operate well below the atmospheric pressure.
   - There is a lower limit to this pressure depending on the temperature of the cooling medium
   - Side effect: Lowering the condenser pressure increases the moisture content of the steam at the final stages of the turbine.

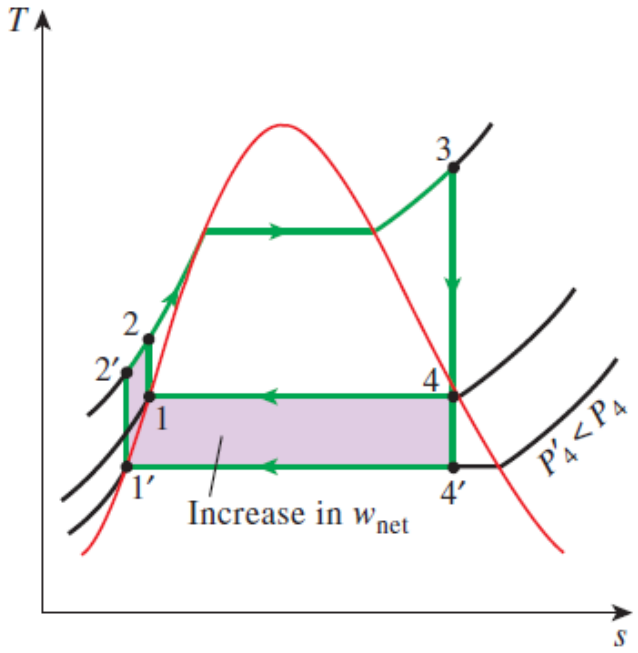


FIGURE 10–6
The effect of lowering the condenser pressure on the ideal Rankine cycle.

10.4 Ways to improve the simple Rankine cycle efficiency (2)

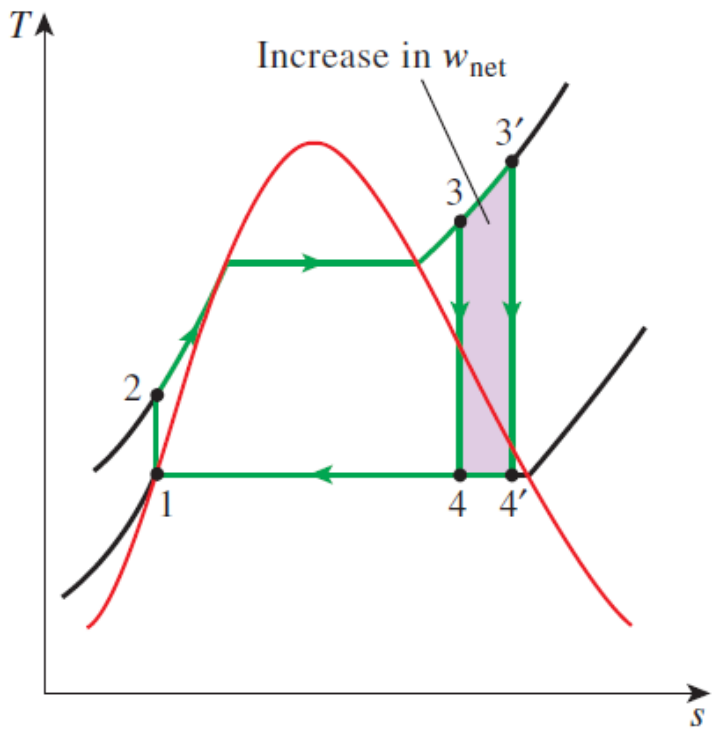


FIGURE 10–7
The effect of superheating the steam to higher temperatures on the ideal Rankine cycle.

2. Superheating the Steam to High Temperatures (*Increases* $T_{high,avg}$)

- Both the net work and heat input increase as a result of superheating the steam to a higher temperature.
- The overall effect is an increase in thermal efficiency since the average temperature at which heat is added increases.
- Superheating to higher temperatures decreases the moisture content of the steam at the turbine exit, which is desirable.
- The temperature is limited by metallurgical considerations.
- Presently the highest steam temperature allowed at the turbine inlet is about 620°C.

10.4 Ways to improve the simple Rankine cycle efficiency (3)

3. Increasing the Boiler Pressure (*Increases* $T_{high,avg}$)

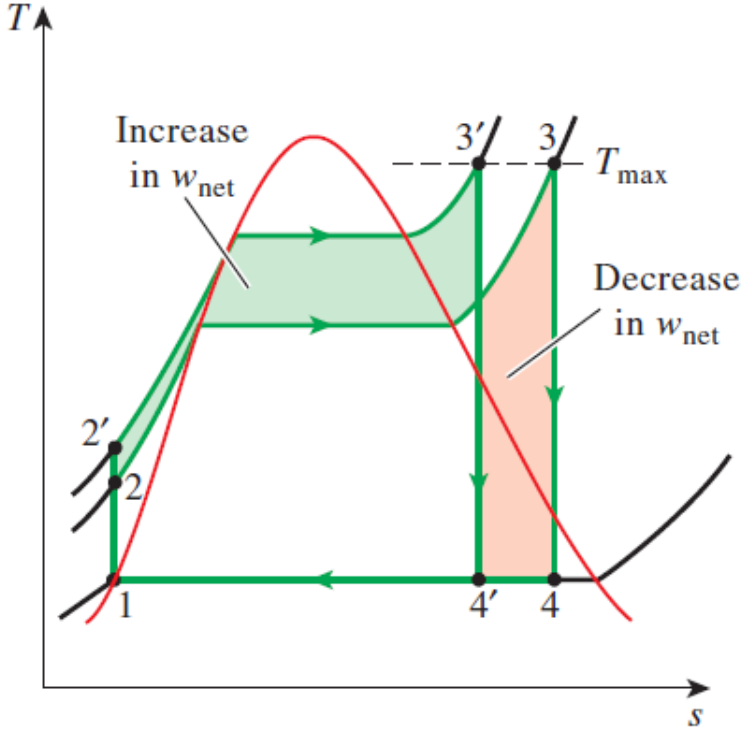


FIGURE 10–8
The effect of increasing the boiler pressure on the ideal Rankine cycle.

- For a fixed turbine inlet temperature, the cycle shifts to the left and the moisture content of steam at the turbine exit increases.
- This side effect can be corrected by reheating the steam.
- Today many modern steam power plants operate at supercritical pressures (P > 22.06 MPa) and have thermal efficiencies of about 40% for fossil-fuel plants and 34% for nuclear plants.

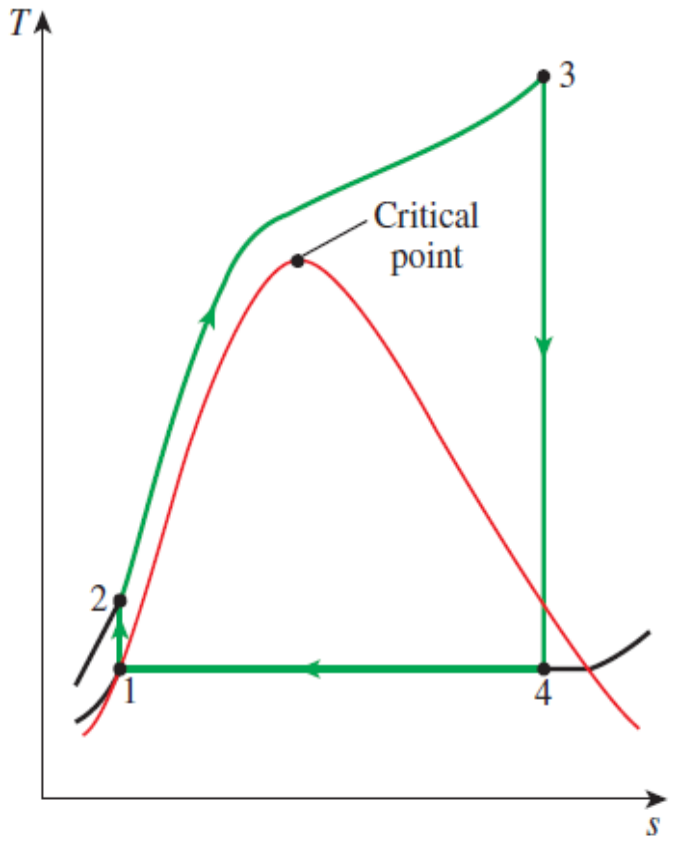


FIGURE 10–9
A supercritical Rankine cycle.

Example 3

EXAMPLE 10–3 **Effect of Boiler Pressure and Temperature on Efficiency**

Consider a steam power plant operating on the ideal Rankine cycle. Steam enters the turbine at 3 MPa and 350°C and is condensed in the condenser at a pressure of 10 kPa. Determine (*a*) the thermal efficiency of this power plant, (*b*) the thermal efficiency if steam is superheated to 600°C instead of 350°C, and (*c*) the thermal efficiency if the boiler pressure is raised to 15 MPa while the turbine inlet temperature is maintained at 600°C.

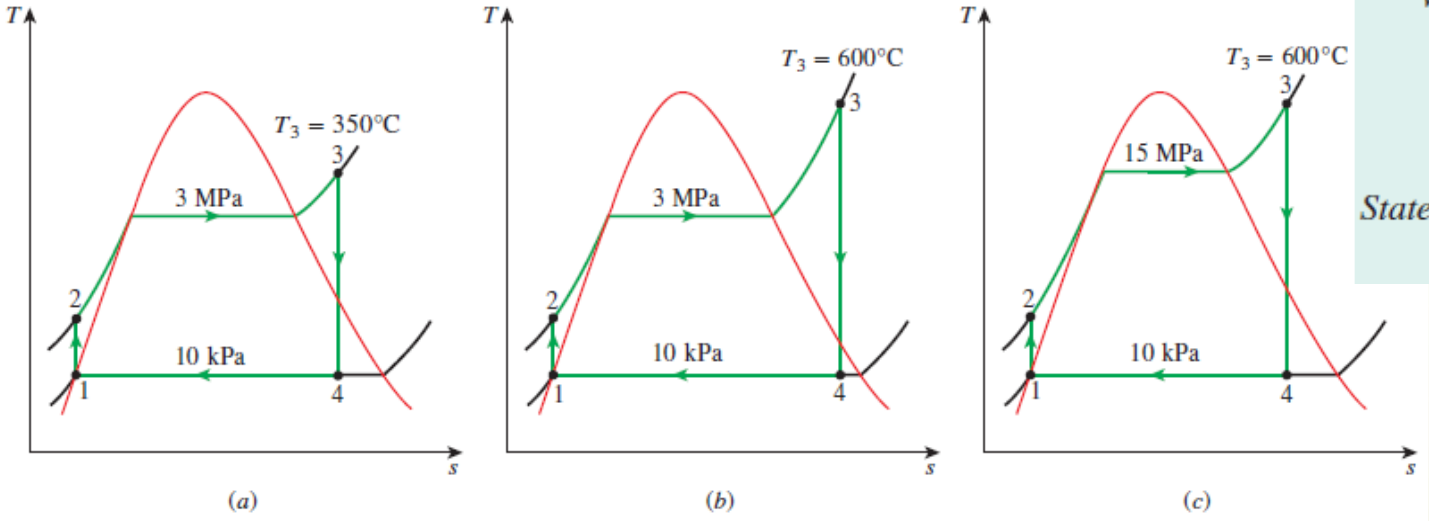


FIGURE 10–10
T-s diagrams of the three cycles discussed in Example 10–3.

SOLUTION A steam power plant operating on the ideal Rankine cycle is considered. The effects of superheating the steam to a higher temperature and raising the boiler pressure on thermal efficiency are to be investigated.

Analysis The *T-s* diagrams of the cycle for all three cases are given in Fig. 10–10. (*a*) This is the steam power plant discussed in Example 10–1, except that the condenser pressure is lowered to 10 kPa. The thermal efficiency is determined in a similar manner:

State 1: $P_1 = 10$ kPa, Sat. liquid $\Big\}$ $h_1 = h_{f\,@\,10\text{ kPa}} = 191.81$ kJ/kg, $v_1 = v_{f\,@\,10\text{ kPa}} = 0.00101$ m³/kg

State 2: $P_2 = 3$ MPa, $s_2 = s_1$

$$w_{\text{pump,in}} = v_1(P_2 - P_1) = (0.00101\ \text{m}^3/\text{kg})[(3000 - 10)\ \text{kPa}]\left(\frac{1\ \text{kJ}}{1\ \text{kPa}\cdot\text{m}^3}\right)$$

$$= 3.02\ \text{kJ/kg}$$

$$h_2 = h_1 + w_{\text{pump,in}} = (191.81 + 3.02)\ \text{kJ/kg} = 194.83\ \text{kJ/kg}$$

State 3: $P_3 = 3$ MPa, $T_3 = 350°\text{C}$ $\Big\}$ $h_3 = 3116.1$ kJ/kg, $s_3 = 6.7450$ kJ/kg·K

State 4: $P_4 = 10$ kPa (sat. mixture)

$$s_4 = s_3$$

$$x_4 = \frac{s_4 - s_f}{s_{fg}} = \frac{6.7450 - 0.6492}{7.4996} = 0.8128$$

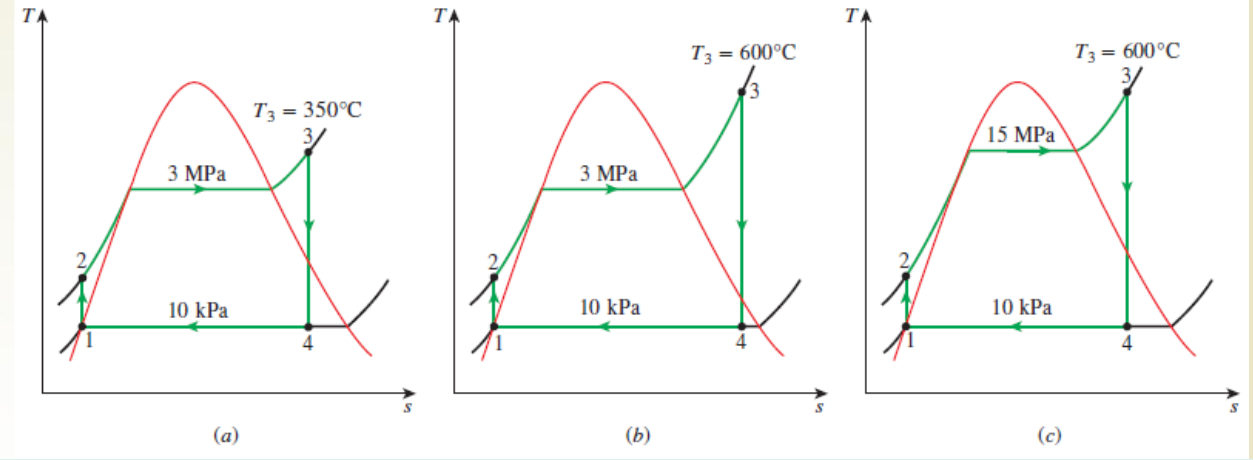
Thus,

$$h_4 = h_f + x_4 h_{fg} = 191.81 + 0.8128(2392.1) = 2136.1 \text{ kJ/kg}$$
$$q_{in} = h_3 - h_2 = (3116.1 - 194.83) \text{ kJ/kg} = 2921.3 \text{ kJ/kg}$$
$$q_{out} = h_4 - h_1 = (2136.1 - 191.81) \text{ kJ/kg} = 1944.3 \text{ kJ/kg}$$

and

$$\eta_{th} = 1 - \frac{q_{out}}{q_{in}} = 1 - \frac{1944.3 \text{ kJ/kg}}{2921.3 \text{ kJ/kg}} = 0.334 \text{ or } \mathbf{33.4\%}$$

Therefore, the thermal efficiency increases from 26.0 to 33.4 percent as a result of lowering the condenser pressure from 75 to 10 kPa. At the same time, however, the quality of the steam decreases from 88.6 to 81.3 percent (in other words, the moisture content increases from 11.4 to 18.7 percent).



(*b*) States 1 and 2 remain the same in this case, and the enthalpies at state 3 (3 MPa and 600°C) and state 4 (10 kPa and $s_4 = s_3$) are determined to be

$$h_3 = 3682.8 \text{ kJ/kg}$$
$$h_4 = 2380.3 \text{ kJ/kg} \quad (x_4 = 0.915)$$

Thus,

$$q_{in} = h_3 - h_2 = 3682.8 - 194.83 = 3488.0 \text{ kJ/kg}$$
$$q_{out} = h_4 - h_1 = 2380.3 - 191.81 = 2188.5 \text{ kJ/kg}$$

and

$$\eta_{th} = 1 - \frac{q_{out}}{q_{in}} = 1 - \frac{2188.5 \text{ kJ/kg}}{3488.0 \text{ kJ/kg}} = 0.373 \text{ or } \mathbf{37.3\%}$$

Therefore, the thermal efficiency increases from 33.4 to 37.3 percent as a result of superheating the steam from 350 to 600°C. At the same time, the quality of the steam increases from 81.3 to 91.5 percent (in other words, the moisture content decreases from 18.7 to 8.5 percent).

(*c*) State 1 remains the same in this case, but the other states change. The enthalpies at state 2 (15 MPa and $s_2 = s_1$), state 3 (15 MPa and 600°C), and state 4 (10 kPa and $s_4 = s_3$) are determined in a similar manner to be

$$h_2 = 206.95 \text{ kJ/kg}$$
$$h_3 = 3583.1 \text{ kJ/kg}$$
$$h_4 = 2115.3 \text{ kJ/kg} \quad (x_4 = 0.804)$$

Thus,

$$q_{in} = h_3 - h_2 = 3583.1 - 206.95 = 3376.2 \text{ kJ/kg}$$
$$q_{out} = h_4 - h_1 = 2115.3 - 191.81 = 1923.5 \text{ kJ/kg}$$

and

$$\eta_{th} = 1 - \frac{q_{out}}{q_{in}} = 1 - \frac{1923.5 \text{ kJ/kg}}{3376.2 \text{ kJ/kg}} = 0.430 \text{ or } \mathbf{43.0\%}$$

Discussion The thermal efficiency increases from 37.3 to 43.0 percent as a result of raising the boiler pressure from 3 to 15 MPa while maintaining the turbine inlet temperature at 600°C. At the same time, however, the quality of the steam decreases from 91.5 to 80.4 percent (in other words, the moisture content increases from 8.5 to 19.6 percent).

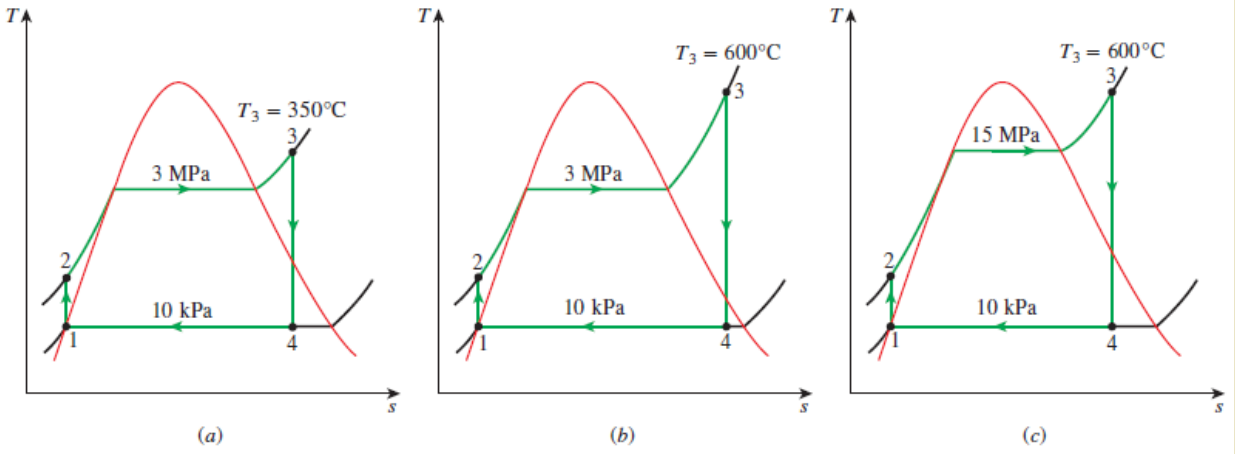


FIGURE 10–10
T-s diagrams of the three cycles discussed in Example 10–3.

10.5 The Ideal Reheat Rankine Cycle

- How can we take advantage of the increased efficiencies at higher boiler pressures without facing the problem of excessive moisture at the final stages of the turbine?
  - Superheat the steam to very high temperatures. It is limited metallurgically.
  - Expand the steam in the turbine in two stages, and reheat it in between (reheat)

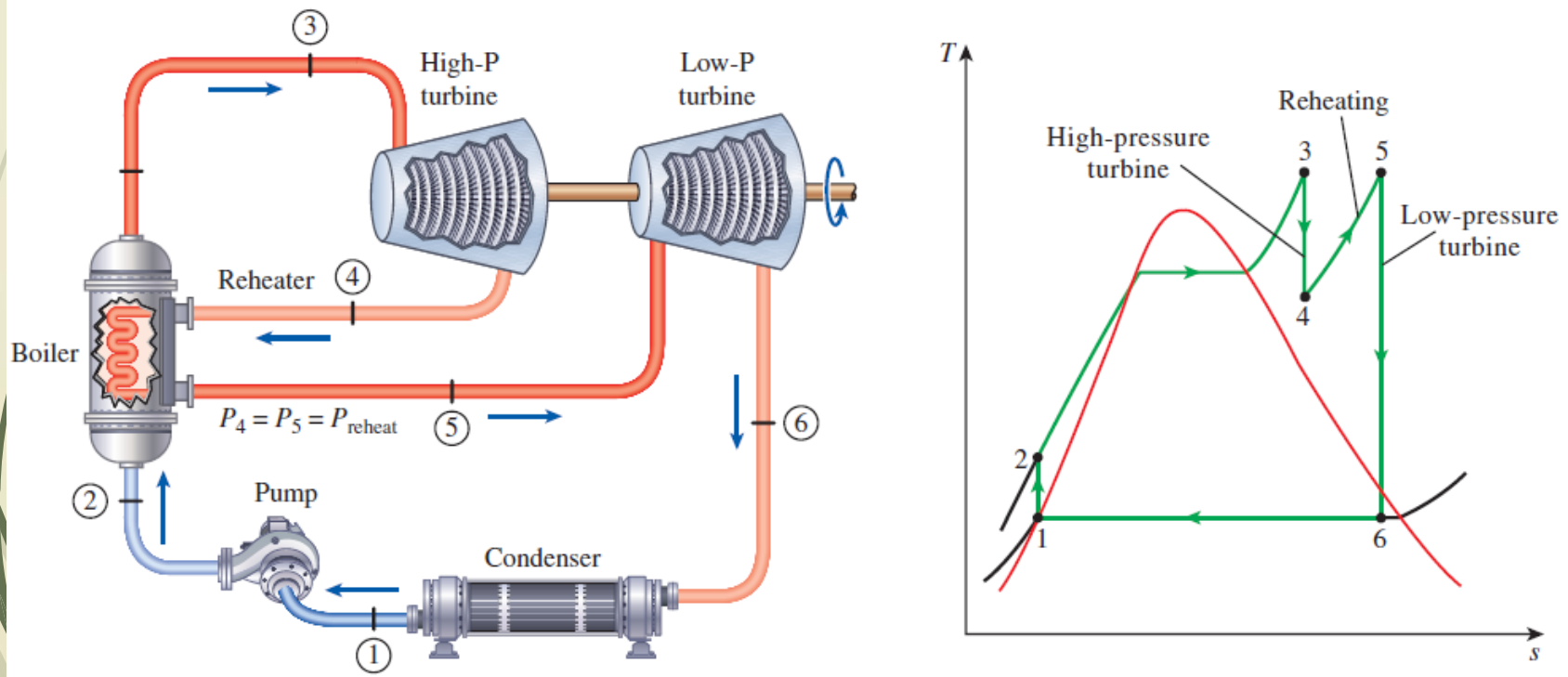


FIGURE 10–11
The ideal reheat Rankine cycle.

10.5 The Ideal Reheat Rankine Cycle (2)

- As the boiler pressure is increased in the simple Rankine cycle, not only does the thermal efficiency increase but also the turbine exit moisture increases.
- The reheat cycle allows the use of higher boiler pressures and provides a means to keep the turbine exit moisture ($x > 0.85$ to 0.90) at an acceptable level.

Component	Process	1st Law Result
1. Boiler	Const. P	$q_{in} = (h_3 - h_2) + (h_5 - h_4)$
2. Turbine	Isentropic	$w_{out} = (h_3 - h_4) + (h_5 - h_6)$
3. Condenser	Const. P	$q_{out} = (h_6 - h_1)$
4. Pump	Isentropic	$w_{in} = (h_2 - h_1) = v_1(P_2 - P_1)$

10.5 The Ideal Reheat Rankine Cycle (3)

- The single reheat in a modern power plant improves the cycle efficiency by 4 to 5% by increasing the average temperature at which heat is transferred to the steam.
- The average temperature during the reheat process can be increased by increasing the number of expansion and reheat stages.
- As the number of stages is increased, the expansion and reheat processes approach an isothermal process at the maximum temperature.
- The use of more than two reheat stages is not practical.
- The theoretical improvement in efficiency from the second reheat is about half of that which results from a single reheat.

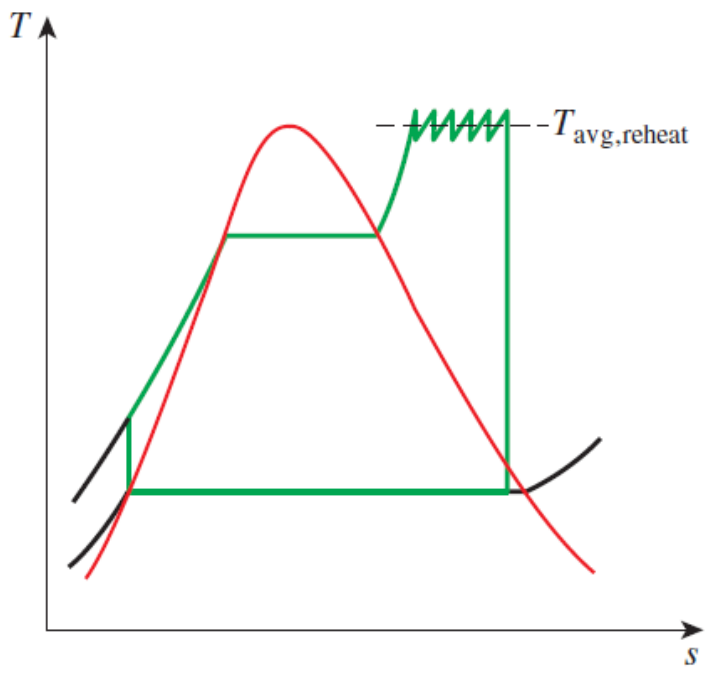


FIGURE 10–12
The average temperature at which heat is transferred during reheating increases as the number of reheat stages is increased.

10.5 The Ideal Reheat Rankine Cycle (3)

- The reheat temperatures are very close or equal to the turbine inlet temperature.
- The optimum reheat pressure is about one-fourth of the maximum cycle pressure.
- The thermal efficiency is given by

$$\eta_{th} = \frac{w_{net}}{q_{in}} = \frac{(h_3 - h_4) + (h_5 - h_6) - (h_2 - h_1)}{(h_3 - h_2) + (h_5 - h_4)}$$

$$= 1 - \frac{h_6 - h_1}{(h_3 - h_2) + (h_5 - h_4)}$$

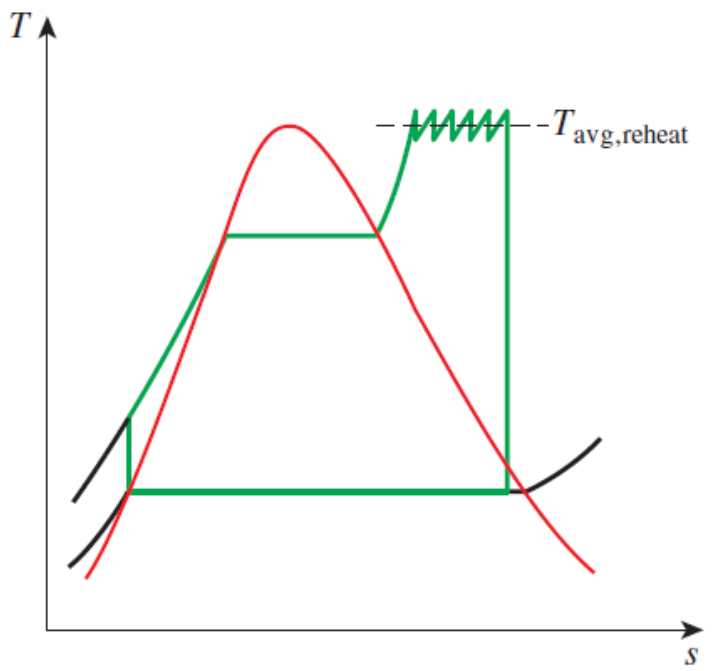


FIGURE 10–12
The average temperature at which heat is transferred during reheating increases as the number of reheat stages is increased.

10.6 The Ideal Regenerative Rankine cycle

- To improve the cycle thermal efficiency, the average temperature at which heat is added must be increased.
- One way to do this is to allow the steam to expand in the turbine to an intermediate pressure. A portion of the steam is extracted from the turbine and sent to a regenerative heater to preheat the condensate before entering the boiler.
- This approach increases the average temperature at which heat is added in the boiler.
- However, this reduces the mass of steam expanding in the lower pressure stages of the turbine; and, thus, the total work done by the turbine. But the work is done efficiently.
- The preheating of the condensate is done in a combination of open and closed feedwater heaters.

10.6 The Ideal Regenerative Rankine cycle (2)

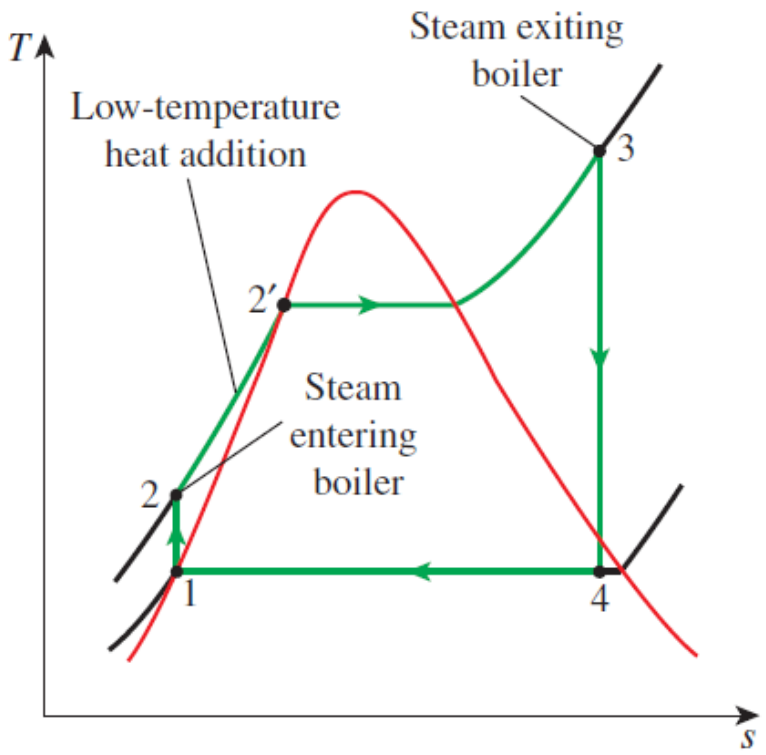


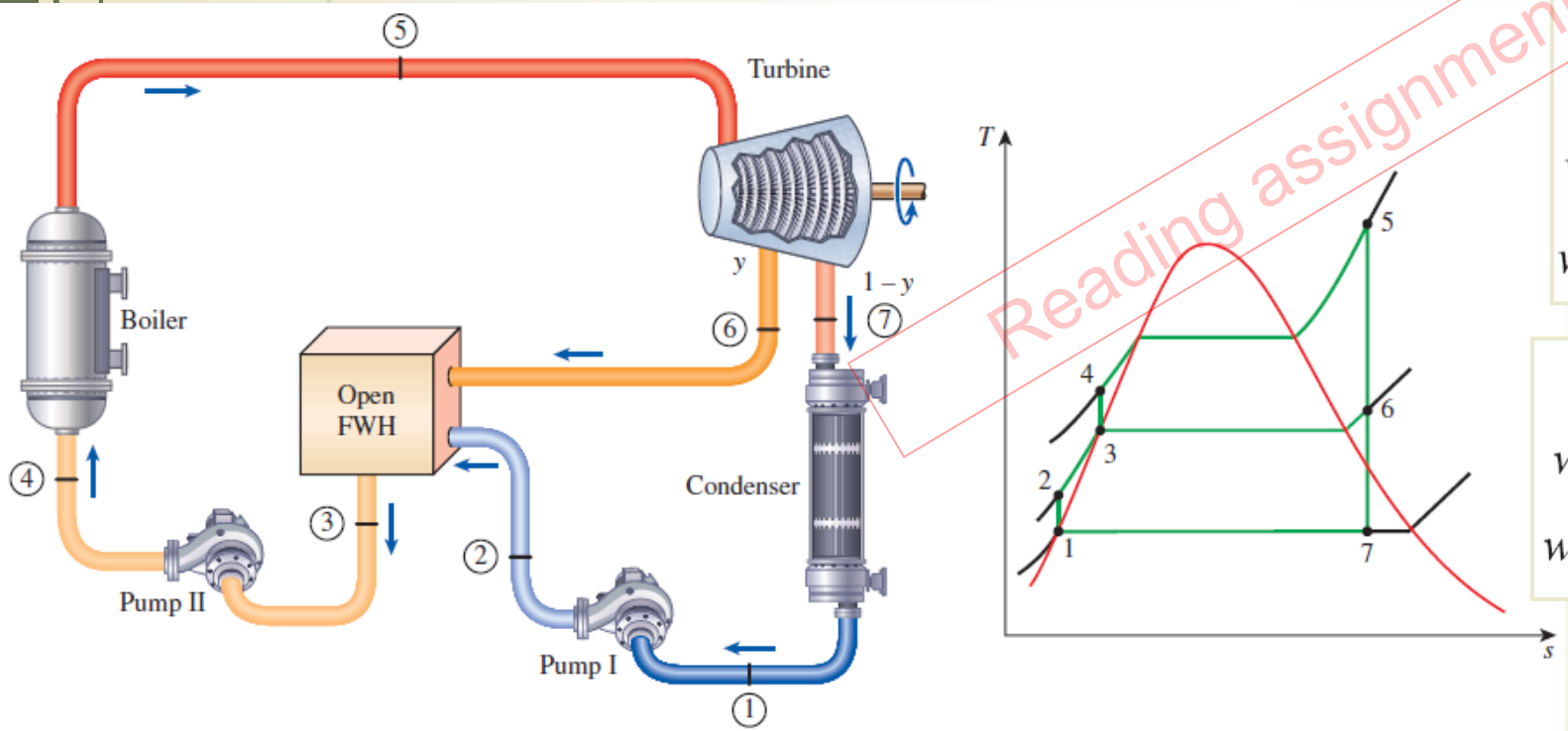
FIGURE 10–14
The first part of the heat-addition process in the boiler takes place at relatively low temperatures.

- Heat is transferred to the working fluid during process 2-2’ at a relatively low temperature.
  - This lowers the average heat-addition temperature and thus the cycle efficiency.
- In steam power plants, steam is extracted from the turbine at various points.
  - This steam, which could have produced more work by expanding further in the turbine, is used to heat the feedwater instead.
- The device where the feedwater is heated by regeneration is called a regenerator, or a feedwater heater (FWH).
  - A feedwater heater is basically a heat exchanger where heat is transferred from the steam to the feedwater either by mixing the two fluid streams (open feedwater heaters) or without mixing them (closed feedwater heaters).

10.6 The Ideal Regenerative Rankine cycle (3)

Open Feedwater Heaters

- An open (or direct-contact) feedwater heater is basically a mixing chamber, where the steam extracted from the turbine mixes with the feedwater exiting the pump.
- Ideally, the mixture leaves the heater as a saturated liquid at the heater pressure.



The ideal regenerative Rankine cycle with an open feedwater heater.

Reading assignment

$$q_{in} = h_5 - h_4$$

$$q_{out} = (1 - y)(h_7 - h_1)$$

$$w_{turb,out} = (h_5 - h_6) + (1 - y)(h_6 - h_7)$$

$$w_{pump,in} = (1 - y)w_{pump\ I,in} + w_{pump\ II,in}$$

$$y = \dot{m}_6/\dot{m}_5 \qquad \text{(fraction of steam extracted)}$$

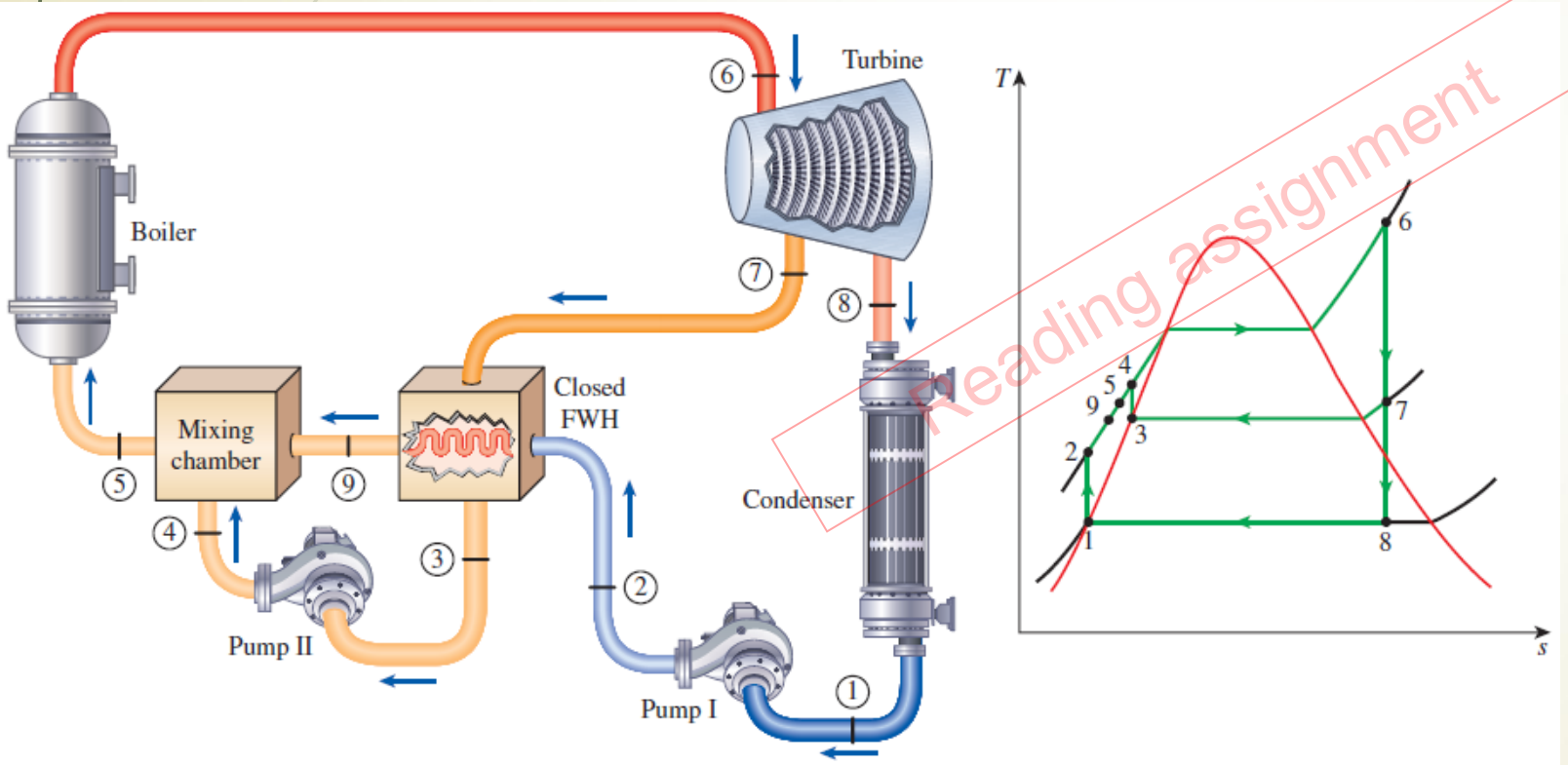
$$w_{pump\ I,in} = v_1(P_2 - P_1)$$

$$w_{pump\ II,in} = v_3(P_4 - P_3)$$

10.6 The Ideal Regenerative Rankine cycle (4)

Closed Feedwater Heaters

- Another type of feedwater heater frequently used in steam power plants is the closed feedwater heater, in which heat is transferred from the extracted steam to the feedwater without any mixing taking place.



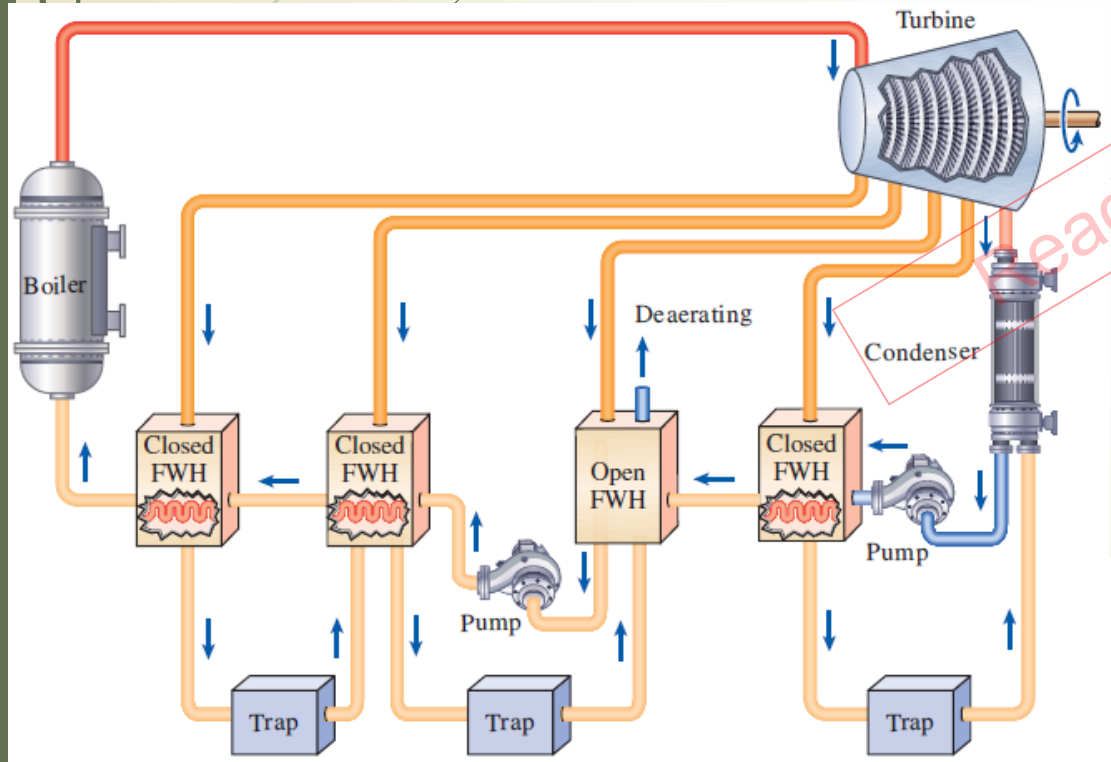
Reading assignment

FIGURE 10–16
The ideal regenerative Rankine cycle with a closed feedwater heater.

- The two streams now can be at different pressures, since they do not mix.

10.6 The Ideal Regenerative Rankine cycle (5)

- The closed feedwater heaters are more complex because of the internal tubing network, and thus they are more expensive.
- Heat transfer in closed feedwater heaters is less effective since the two streams are not allowed to be in direct contact.
- However, closed feedwater heaters do not require a separate pump for each heater since the extracted steam and the feedwater can be at different pressures.



- Open feedwater heaters are simple and inexpensive and have good heat transfer characteristics.
  - For each heater, however, a pump is required to handle the feedwater.
- Most steam power plants use a combination of open and closed feedwater heaters.

FIGURE 10–17
A steam power plant with one open and three closed feedwater heaters.

Reading assignment

10.8 Cogeneration

- Many industries require energy input in the form of heat, called process heat.
  - Process heat in these industries is usually supplied by steam at 5 to 7 atm and 150 to 200°C.
  - Energy is usually transferred to the steam by burning coal, oil, natural gas, or another fuel in a furnace.

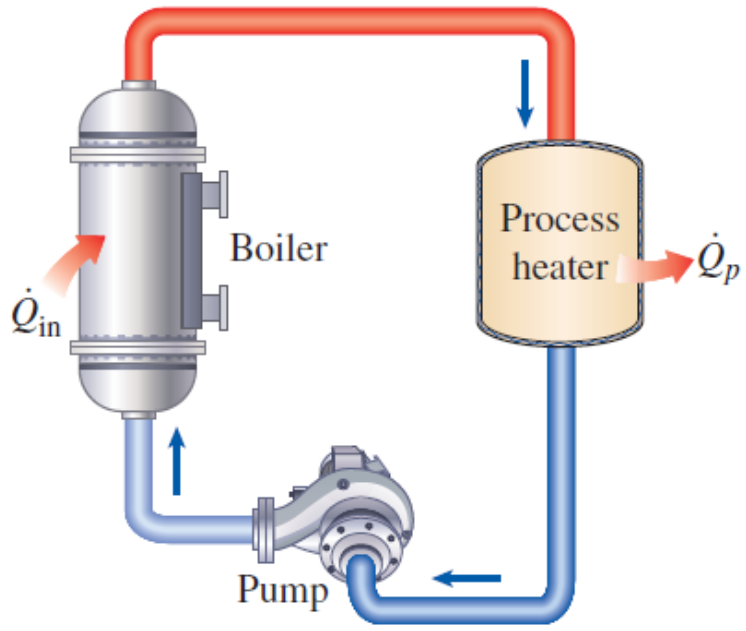


FIGURE 10–21
A simple process-heating plant.

- Industries that use large amounts of process heat also consume a large amount of electric power.
- It makes sense to use the already-existing work potential to produce power instead of letting it go to waste.
- The result is a plant that produces electricity while meeting the process-heat requirements of certain industrial processes (cogeneration plant)
- Cogeneration: The production of more than one useful form of energy (such as process heat and electric power) from the same energy source.

10.8 Cogeneration (2)

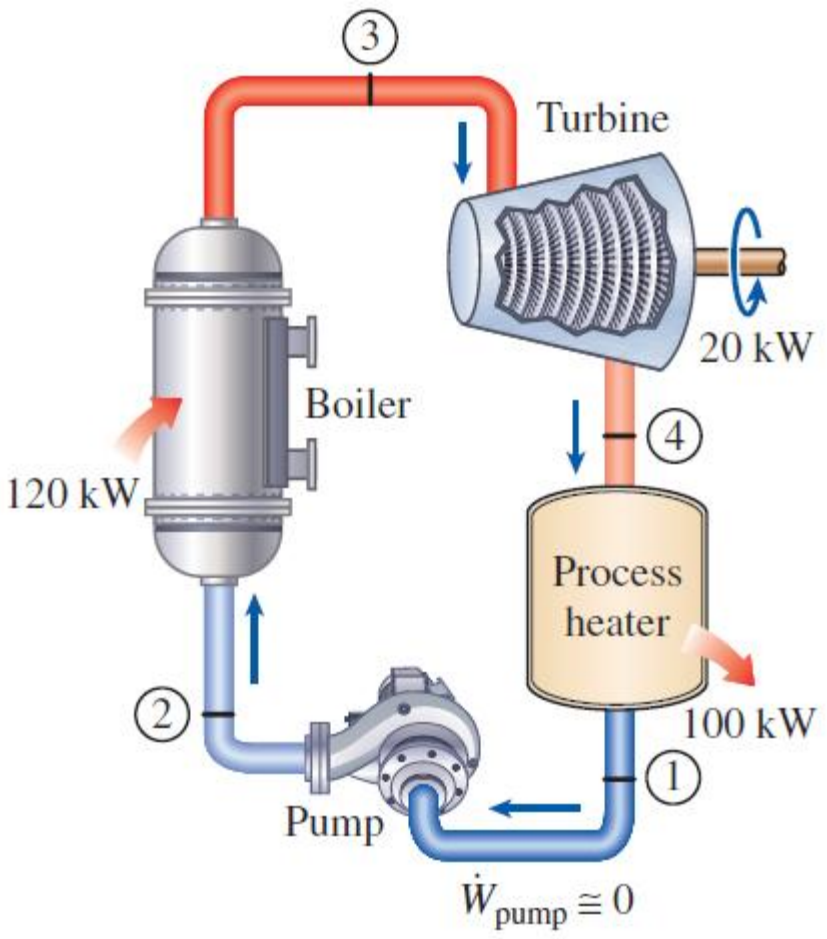


FIGURE 10–22
An ideal cogeneration plant.

Utilization factor

$$\epsilon_u = \frac{\text{Net power output} + \text{Process heat delivered}}{\text{Total heat input}} = \frac{\dot{W}_{net} + \dot{Q}_p}{\dot{Q}_{in}}$$

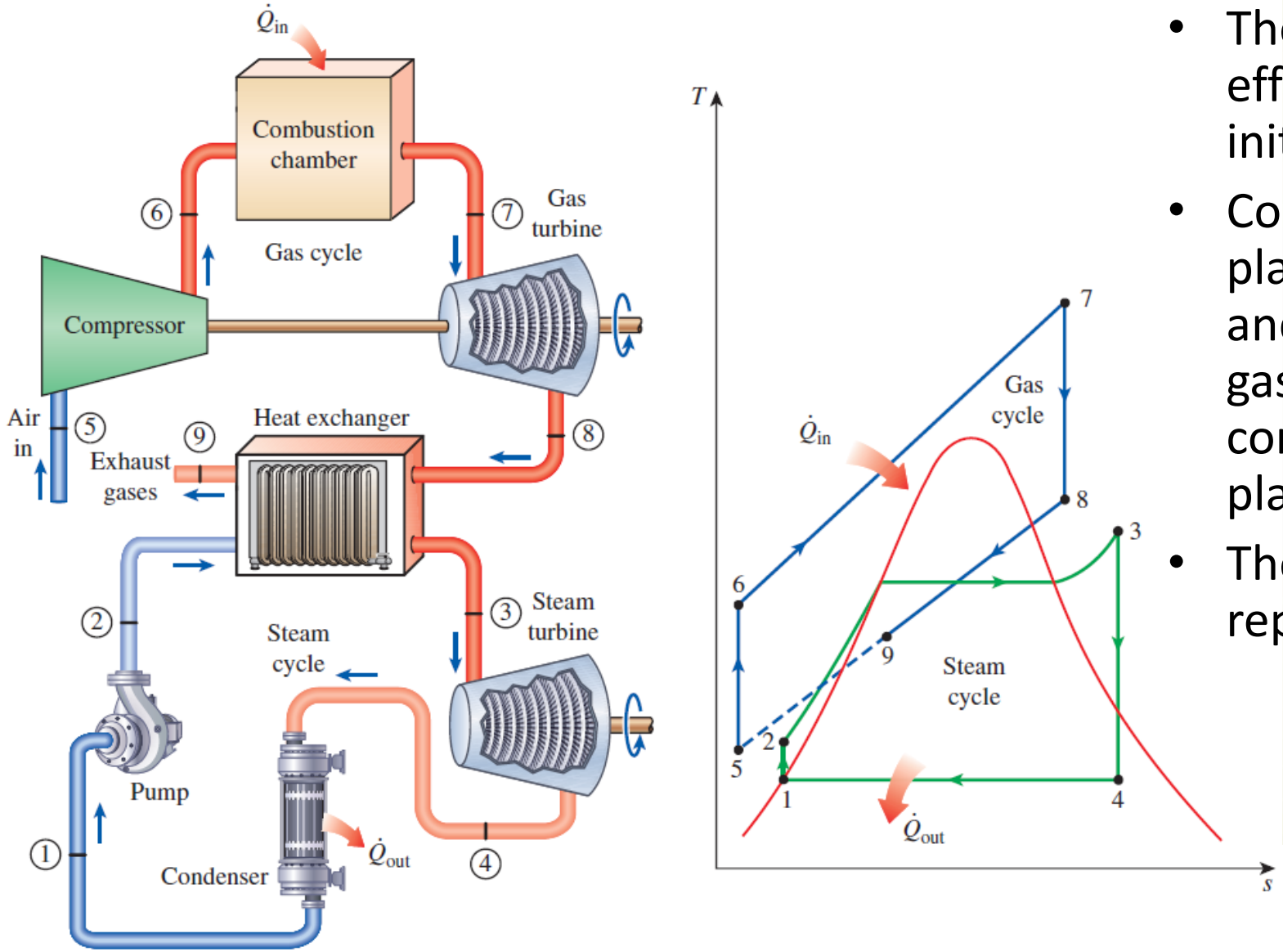
$$\epsilon_u = 1 - \frac{\dot{Q}_{out}}{\dot{Q}_{in}}$$

- The utilization factor of the ideal steam-turbine cogeneration plant is 100%.
- Actual cogeneration plants have utilization factors as high as 80%.
- Some recent cogeneration plants have even higher utilization factors.

10.9 Combined Gas-Vapor Cycles

- The continued quest for higher thermal efficiencies has resulted in rather innovative modifications to conventional power plants.
- A popular modification involves a gas power cycle topping a vapor power cycle, which is called the combined gas–vapor cycle, or just the combined cycle.
- The combined cycle of greatest interest is the gas-turbine (Brayton) cycle topping a steam-turbine (Rankine) cycle, which has a higher thermal efficiency than either of the cycles executed individually.
- It makes engineering sense to take advantage of the very desirable characteristics of the gas-turbine cycle at high temperatures and to use the high-temperature exhaust gases as the energy source for the bottoming cycle such as a steam power cycle. The result is a combined gas–steam cycle.
- Recent developments in gas-turbine technology have made the combined gas–steam cycle economically very attractive.

10.9 Combined Gas-Vapor Cycles (2)



- The combined cycle increases the efficiency without increasing the initial cost greatly.
- Consequently, many new power plants operate on combined cycles, and many more existing steam- or gas-turbine plants are being converted to combined-cycle power plants.
- Thermal efficiencies over 50% are reported.

FIGURE 10–25
Combined gas–steam power plant.

Example

EXAMPLE 10–9 A Combined Gas–Steam Power Cycle

Consider the combined gas–steam power cycle shown in Fig. 10–27. The topping cycle is a gas-turbine cycle that has a pressure ratio of 8. Air enters the compressor at 300 K and the turbine at 1300 K. The isentropic efficiency of the compressor is 80 percent, and that of the gas turbine is 85 percent. The bottoming cycle is a simple ideal Rankine cycle operating between the pressure limits of 7 MPa and 5 kPa. Steam is heated in a heat exchanger by the exhaust gases to a temperature of 500°C. The exhaust gases leave the heat exchanger at 450 K. Determine (*a*) the ratio of the mass flow rates of the steam and the combustion gases and (*b*) the thermal efficiency of the combined cycle.

SOLUTION A combined gas–steam cycle is considered. The ratio of the mass flow rates of the steam and the combustion gases and the thermal efficiency are to be determined.

Analysis The *T-s* diagrams of both cycles are given in Fig. 10–27. The gas-turbine cycle alone was analyzed in Example 9–6, and the steam cycle in Example 10–8*b*, with the following results:

Gas cycle:

$$h_{4'} = 880.36 \text{ kJ/kg} \quad (T_{4'} = 853 \text{ K})$$
$$q_{in} = 790.58 \text{ kJ/kg} \quad w_{net} = 210.41 \text{ kJ/kg} \quad \eta_{th} = 26.6\%$$
$$h_{5'} = h_{@\,450\text{ K}} = 451.80 \text{ kJ/kg}$$

Steam cycle:

$$h_2 = 144.78 \text{ kJ/kg} \quad (T_2 = 33°\text{C})$$
$$h_3 = 3411.4 \text{ kJ/kg} \quad (T_3 = 500°\text{C})$$
$$w_{net} = 1331.4 \text{ kJ/kg} \quad \eta_{th} = 40.8\%$$

(*a*) The ratio of mass flow rates is determined from an energy balance on the heat exchanger:

$$\dot{E}_{in} = \dot{E}_{out}$$
$$\dot{m}_g h_{5'} + \dot{m}_s h_3 = \dot{m}_g h_{4'} + \dot{m}_s h_2$$
$$\dot{m}_s(h_3 - h_2) = \dot{m}_g(h_{4'} - h_{5'})$$
$$\dot{m}_s(3411.4 - 144.78) = \dot{m}_g(880.36 - 451.80)$$

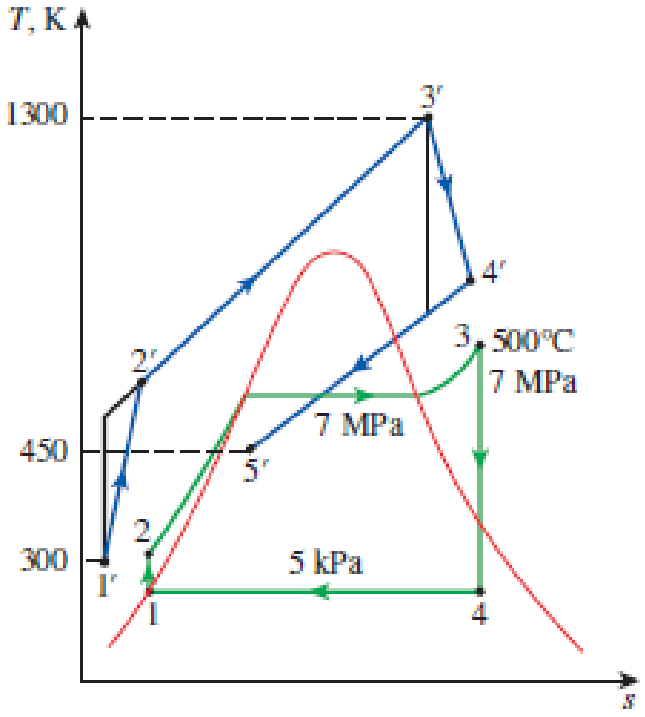


FIGURE 10–27
T-s diagram of the gas–steam combined cycle described in Example 10–9.

Thus,

$$\frac{\dot{m}_s}{\dot{m}_g} = y = \mathbf{0.131}$$

That is, 1 kg of exhaust gases can heat only 0.131 kg of steam from 33 to 500°C as they are cooled from 853 to 450 K. Then the total net work output per kilogram of combustion gases becomes

$$\begin{aligned} w_{net} &= w_{net,gas} + y w_{net,steam} \\ &= (210.41 \text{ kJ/kg gas}) + (0.131 \text{ kg steam/kg gas})(1331.4 \text{ kJ/kg steam}) \\ &= 384.8 \text{ kJ/kg gas} \end{aligned}$$

Therefore, for each kg of combustion gases produced, the combined plant will deliver 384.8 kJ of work. The net power output of the plant is determined by multiplying this value by the mass flow rate of the working fluid in the gas-turbine cycle.

(*b*) The thermal efficiency of the combined cycle is determined from

$$\eta_{th} = \frac{w_{net}}{q_{in}} = \frac{384.8 \text{ kJ/kg gas}}{790.6 \text{ kJ/kg gas}} = 0.487 \text{ or } \mathbf{48.7\%}$$

Discussion Note that this combined cycle converts to useful work 48.7 percent of the energy supplied to the gas in the combustion chamber. This value is considerably higher than the thermal efficiency of the gas-turbine cycle (26.6 percent) or the steam-turbine cycle (40.8 percent) operating alone.

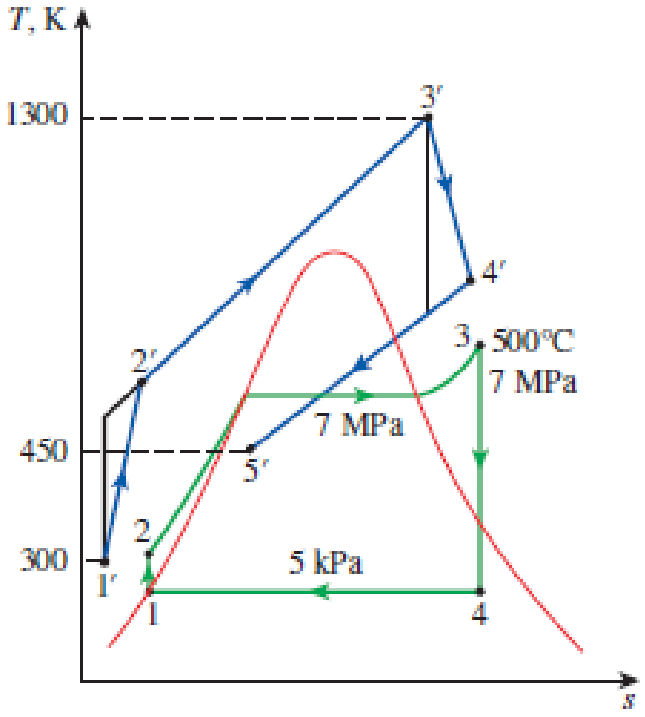


FIGURE 10–27
T-s diagram of the gas–steam combined cycle described in Example 10–9.

Thank You